

SAMSUNG



Innovation Campus

Economic Impact Analysis of Global Food Wastage Analysis



Together for Tomorrow!
Enabling People
Education for Future Generations

Background

- This project uses Machine Learning to analyze and predict the economic losses caused by global food waste. Using data from 2018 to 2024, we transitioned from basic demographic correlations to a sophisticated risk-classification model that identifies supply chain vulnerabilities.
- Food wastage represents a systemic failure in the global supply chain, leading to billions of dollars in preventable financial losses. While traditional systems focus on descriptive reporting, our approach utilizes predictive AI to enable proactive interventions.
- Food waste is also an economic crisis. It contributes to financial instability and resource exhaustion. By predicting "high-risk" zones using economic and climatic indicators, we provide a data-driven framework for supply chain efficiency.

Objectives

- The primary goal is to design a predictive framework that minimizes the financial and environmental costs of food waste through data-driven forecasting.

Machine Learning Goals:

- To eliminate demographic bias by focusing on per-capita waste dynamics.
- To quantify the impact of macroeconomic factors and environmental shifts on waste patterns.
- To develop a Risk Classification Model (High vs. Low Risk) that identifies regions requiring urgent policy or supply chain intervention.
- To transition from high-error linear models to powerful non-linear Ensemble Methods for better generalizability.

SDG Relation

- **SDG 12 – Responsible Consumption and Production:** Our project is a direct implementation tool for Target 12.3, which aims to **halve global per capita food waste** at the retail and consumer levels by 2030.
- **Economic Stability SDG 8:** By reducing financial leakage in the supply chain, the project supports sustainable economic growth.
- **Climate Action SDG 13:** By integrating temperature change as a feature, we highlight the link between climate volatility and food spoilage, promoting a more resilient cold chain.
- Our model identifies future risk zones, allowing for better inventory management and consumer awareness, instead of reacting to past waste data.

Data

- Primary Data: **Global Food Wastage Dataset (2018-2024)** containing 5,000+ records.
- Exogenous Variables: **Food Inflation Rate** (YoY %) from the World Bank and **Average Annual Temperature Change** (°C) from FAOSTAT.

Preprocessing Steps:

- **Feature Engineering:** Transformed the target variable from "Total Economic Loss" to "Food Waste Per Capita (KG)" to eliminate population-driven target leakage.
- **Outlier Detection:** Implemented Isolation Forest to detect and remove extreme historical anomalies that would skew model boundaries.
- **Normalization:** Merged disparate economic and climatic data using dynamic 'Country' and 'Year' indices.
- **Feature Selection:** Manually eliminated absolute scale variables (Population, Total Tonnage) to force the model to learn behavioral patterns rather than demographic sizes.

Model Selection and Training/Testing

SAMSUNG



- Our initial model suffered from target leakage because it relied on absolute population mass, resulting in a misleading 0.99 score. To fix this, we moved from simple demographics to real socioeconomic insights. First, we applied Per-Capita Normalization by shifting our target from 'Total Waste' to 'Waste Per Person' to eliminate population bias. Second, we integrated Exogenous Variables—Food Inflation from the World Bank and Temperature Change from FAOSTAT—to capture real-world economic and climate data. Lastly, we used an Isolation Forest to clean out localized historical crises before training, and switched to a Random Forest Classifier to handle the non-linear relationships between these features.
- When comparing our models, you will notice a drop in accuracy from 96.88% to 54.72%. In data science, this is actually a major victory. The previous high score was because the model was simply memorizing population sizes and geographic borders. By removing that target leakage, our new metrics are highly realistic and well-balanced.



Deployment

Transitioning our static model into a secure, live FastAPI microservice (API) that generates real-time risk predictions using exogenous climate and food inflation data.

- Model Serialization
- Model Serving & API Integration
- Security Considerations
- Monitoring and Logging

Model Serialization

SAMSUNG



Bundling the finalized Random Forest model, feature metadata, and the operational median threshold into a single binary file using joblib. This ensures consistent risk classification boundaries and strictly prevents data leakage during live inferences.

Model Serving & API Integration

Exposing the serialized model as a high-performance microservice using FastAPI. The synchronous POST endpoint processes incoming JSON inputs (inflation and climate data) to deliver instant risk classifications and model confidence scores in milliseconds.

Security Considerations

SAMSUNG



Enforcing strict input validation and type-safety with Pydantic to block malformed payloads at the API gateway, combined with rate-limiting protocols to defend the live microservice against Denial-of-Service (DoS) attacks.

Monitoring and Logging

Establishing a structured logging pipeline with Python's logging framework to track real-time inference latency, audit input data drift, and isolate runtime errors without exposing system internals.

Evaluation Results

SAMSUNG



In conclusion,

- Target Leakage Eliminated: The drop in apparent accuracy is the mathematical proof that our model stopped blindly memorizing total country population.
- Balanced Stability: The clean convergence of Precision (56.48%), Recall (52.44%), and F1-Score (54.38%) confirms a stable model that accurately isolates real risks without favoring one class.
- A Realistic Baseline: Exogenous factors like food inflation and local climate change establish a solid macroeconomic foundation for risk zones, proving that complex behavioral predictions require sound feature engineering rather than inflated numbers
- Resolving Overfitting: Eliminating the population bias by successfully transitioning to a robust, per-capita risk classification model.

SAMSUNG



Innovation Campus

Thank you!



Together for Tomorrow!
Enabling People
Education for Future Generations